

The Network Layer: A Datagram's Journey

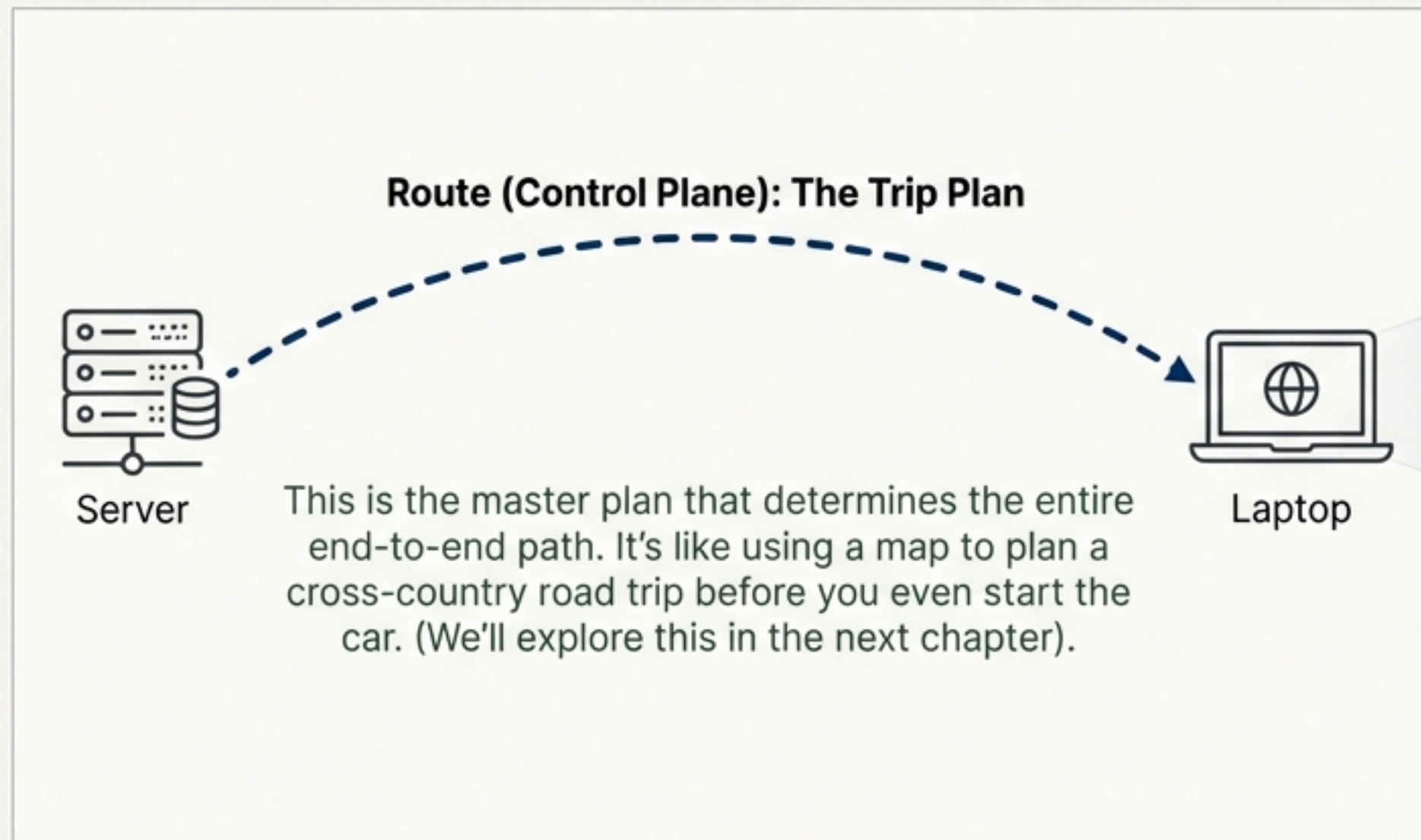
An inside look at the Data Plane—how packets find their way across the internet.



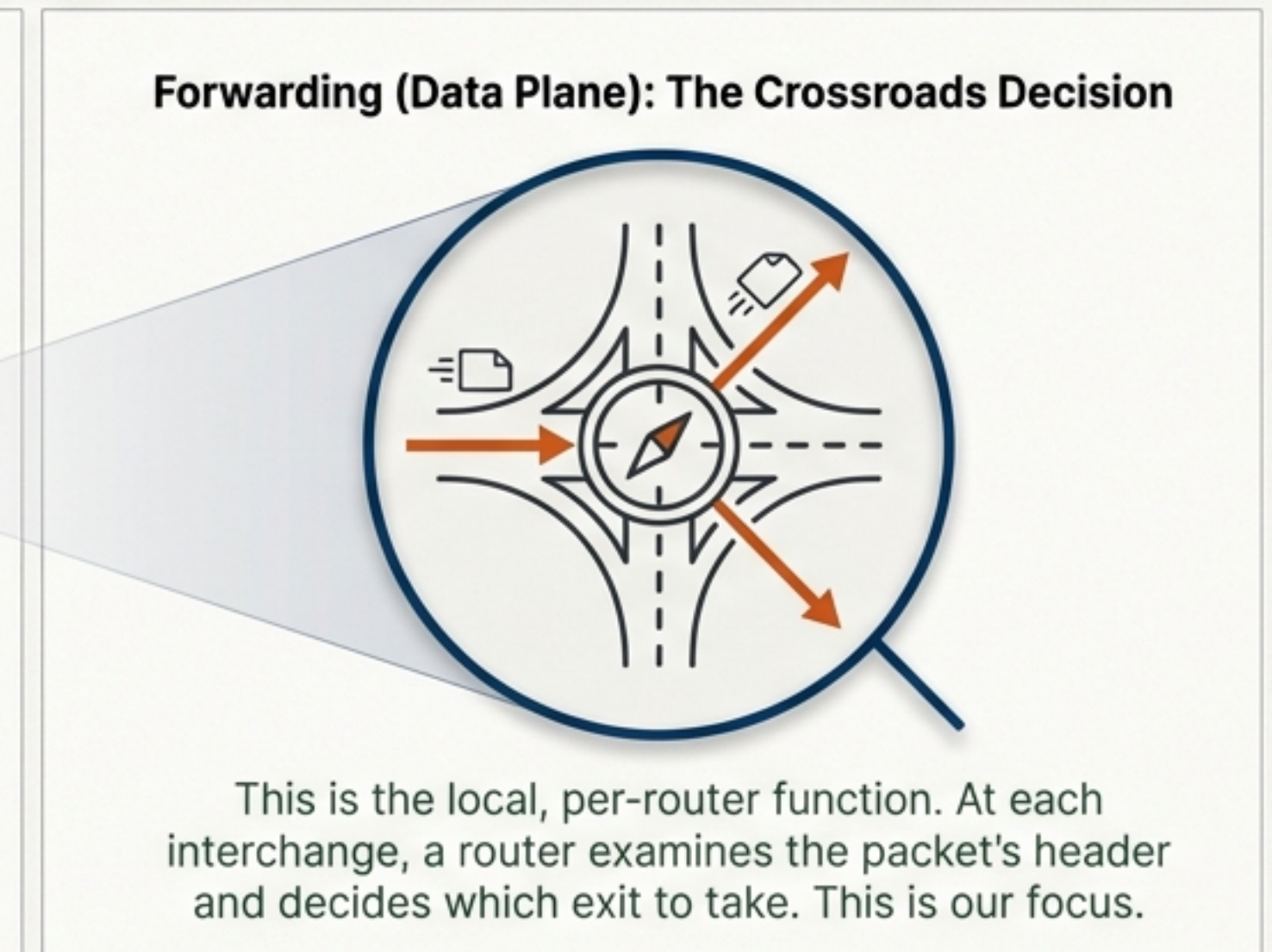
The Mission: From Server to You

Our story follows a single datagram traveling from a web server to your laptop. The Network Layer is its travel agent and its local traffic cop.

Part 1: The Trip Plan (Routing)



Part 2: The Crossroads Decision (Forwarding)



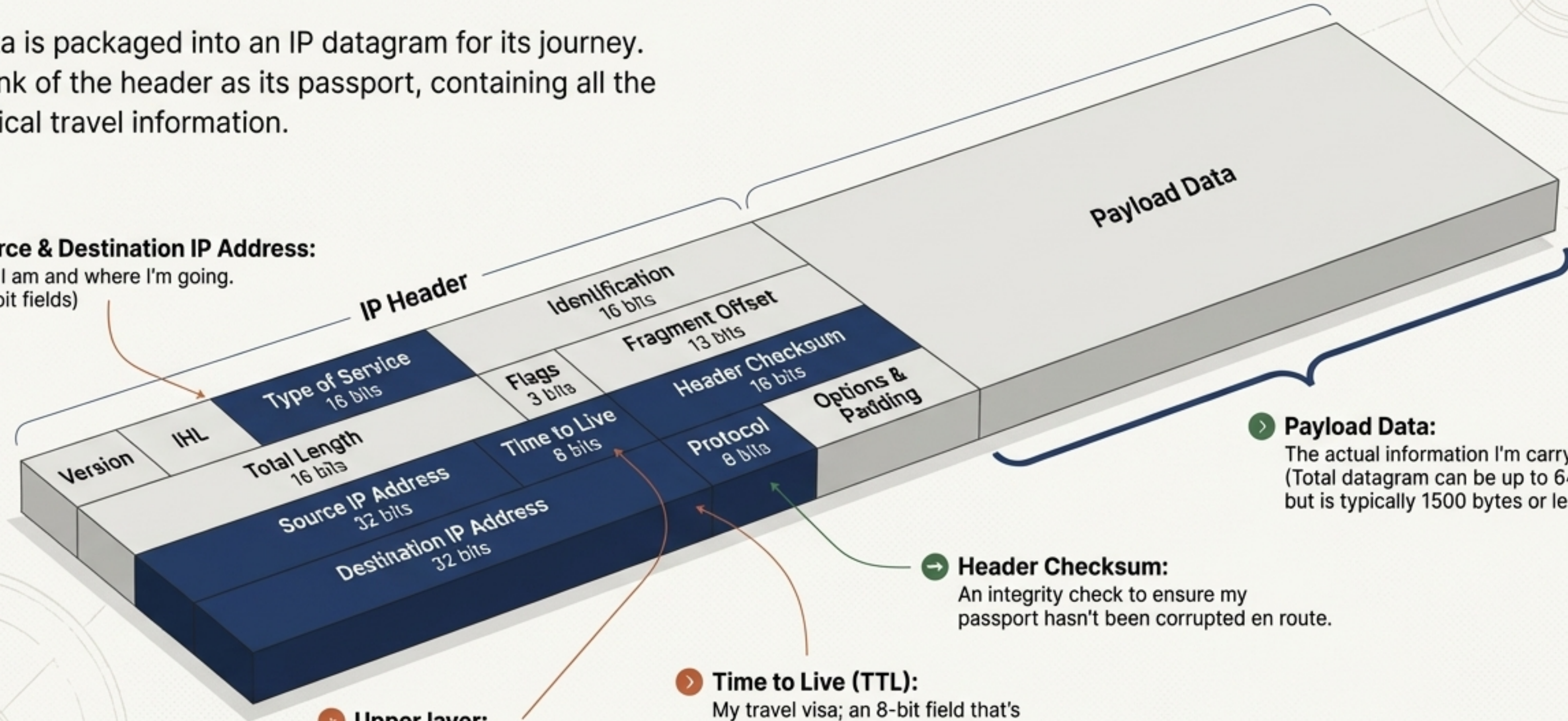
The Traveler: Anatomy of an IP Datagram

Data is packaged into an IP datagram for its journey. Think of the header as its passport, containing all the critical travel information.



Source & Destination IP Address:

Who I am and where I'm going.
(32-bit fields)



Payload Data:

The actual information I'm carrying.
(Total datagram can be up to 64K bytes, but is typically 1500 bytes or less).



Header Checksum:

An integrity check to ensure my passport hasn't been corrupted en route.



Time to Live (TTL):

My travel visa; an 8-bit field that's decremented at each router to prevent me from wandering forever.

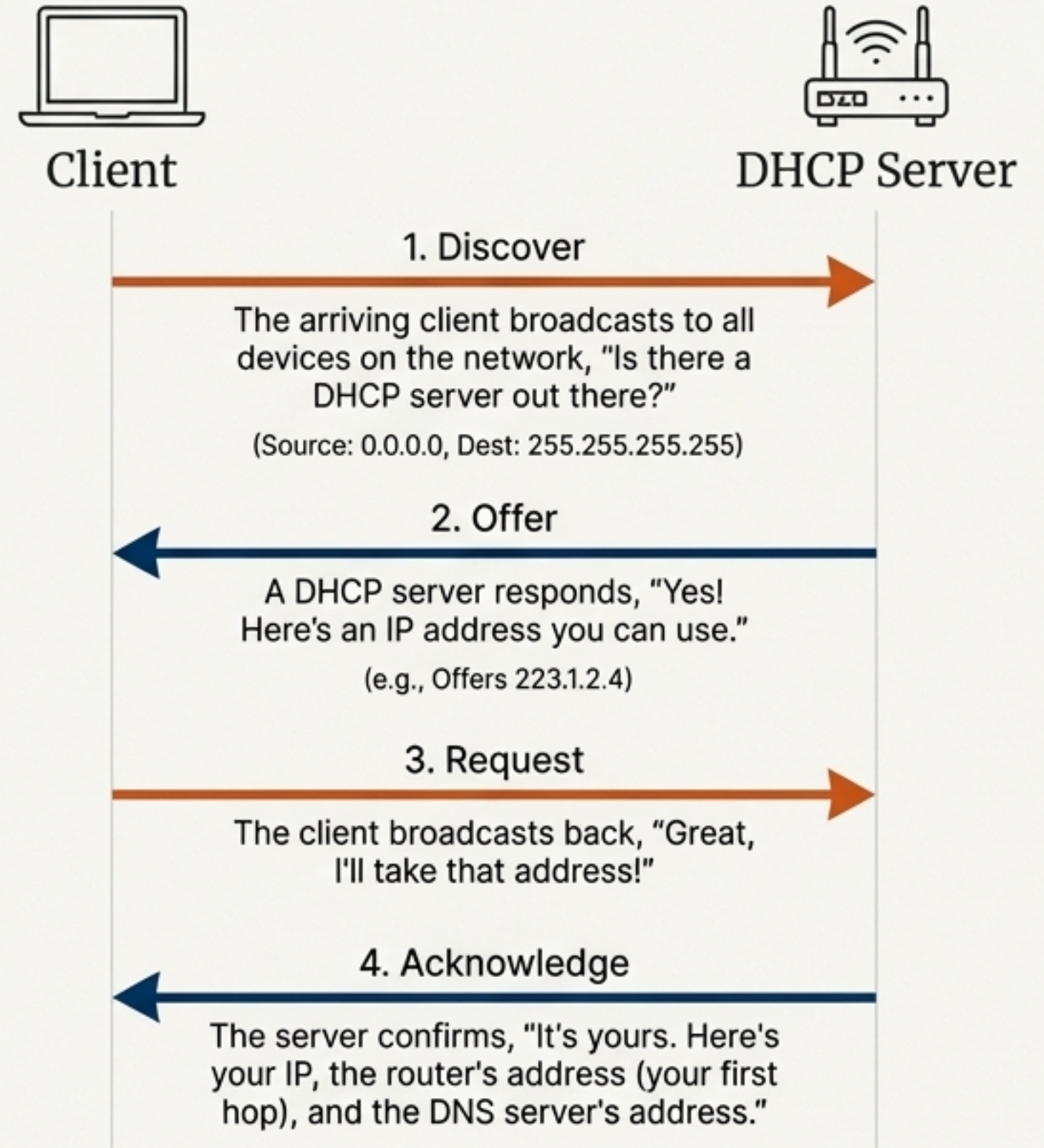


Upper layer:

My 8-bit field telling the destination what kind of payload I'm carrying (e.g., TCP or UDP).

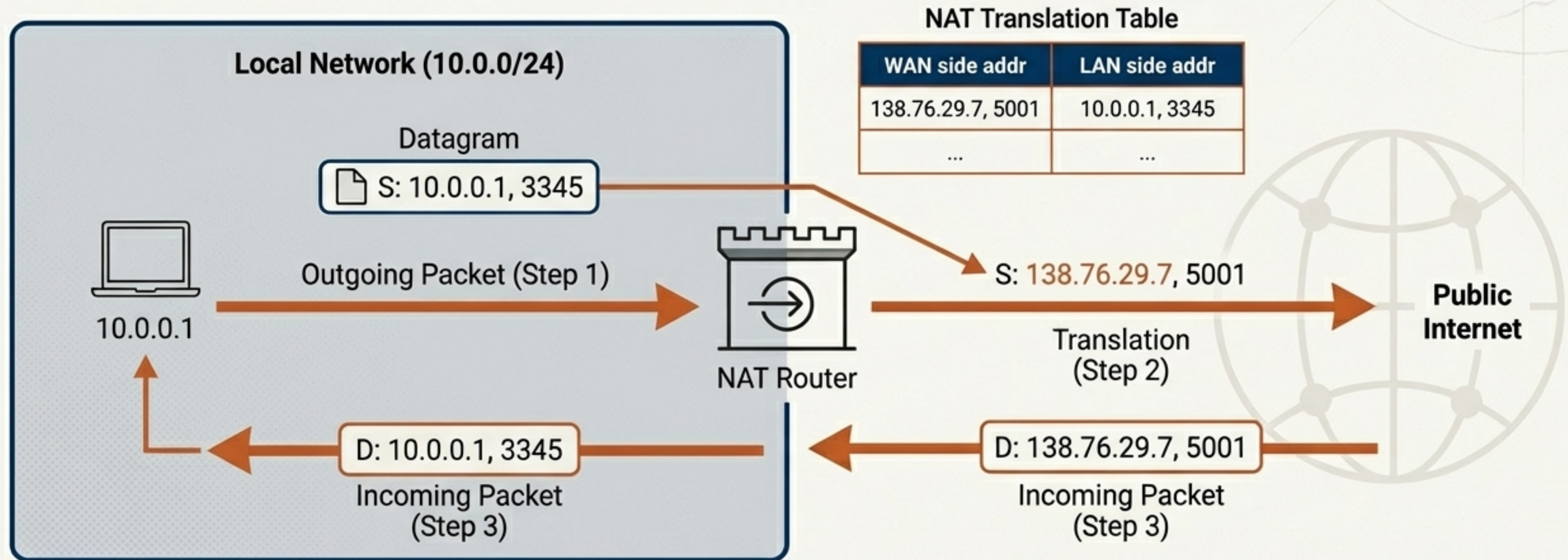
Starting the Journey: The Address is Everything

Every device needs a unique IP address to communicate. When your laptop joins a new network, it can't just pick one; it has to ask for one. This is handled by DHCP, the network's front desk.



Leaving Home: A Change of Identity at the NAT Gateway

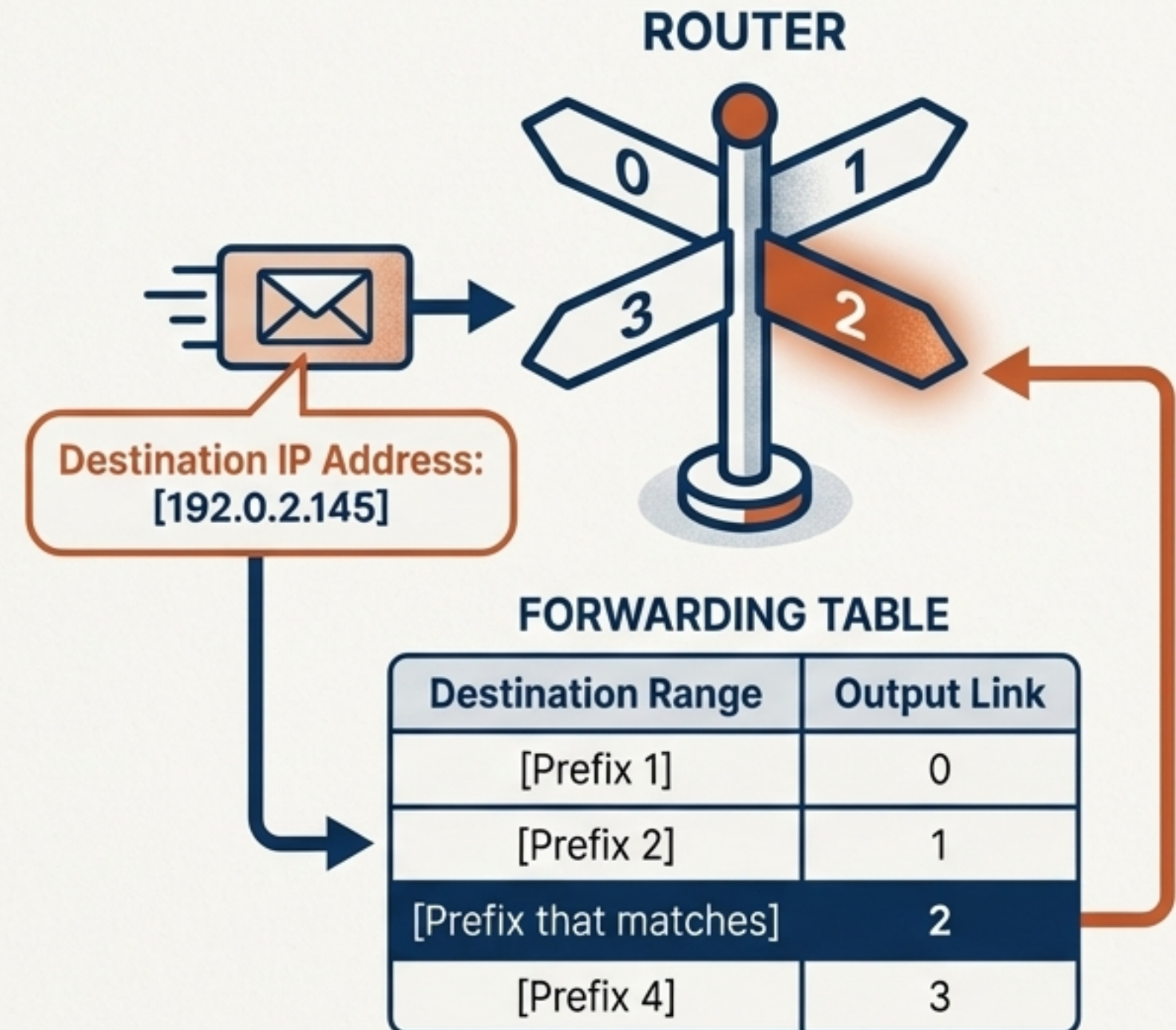
Datagrams inside a local network use private IP addresses (e.g., 10.0.0.1). To the outside world, the entire network appears as a single, public IP address. The NAT router acts as the doorman, managing this translation.



The Crossroads: A Router's Forwarding Decision

The datagram arrives at a router's input port. The router's job is simple: examine the destination IP address and forward the datagram to the correct output port to get it one step closer to its final destination.

To do this, it consults its forwarding table. This table is a set of rules, like signposts, that map destination address ranges to specific output links. The decision is local and must be made in nanoseconds.



Reading the Signs: The Art of Longest Prefix Matching

A single destination address might match multiple entries in the forwarding table. For example, a route for a specific organization and a general “everything else” route.

The rule: The router always chooses the *most specific* route. This is the **Longest Prefix Match**. It finds the entry in its table where the prefix has the longest number of matching bits with the datagram's destination address.

Destination Address: 11001000 00010111 00010110 10100001

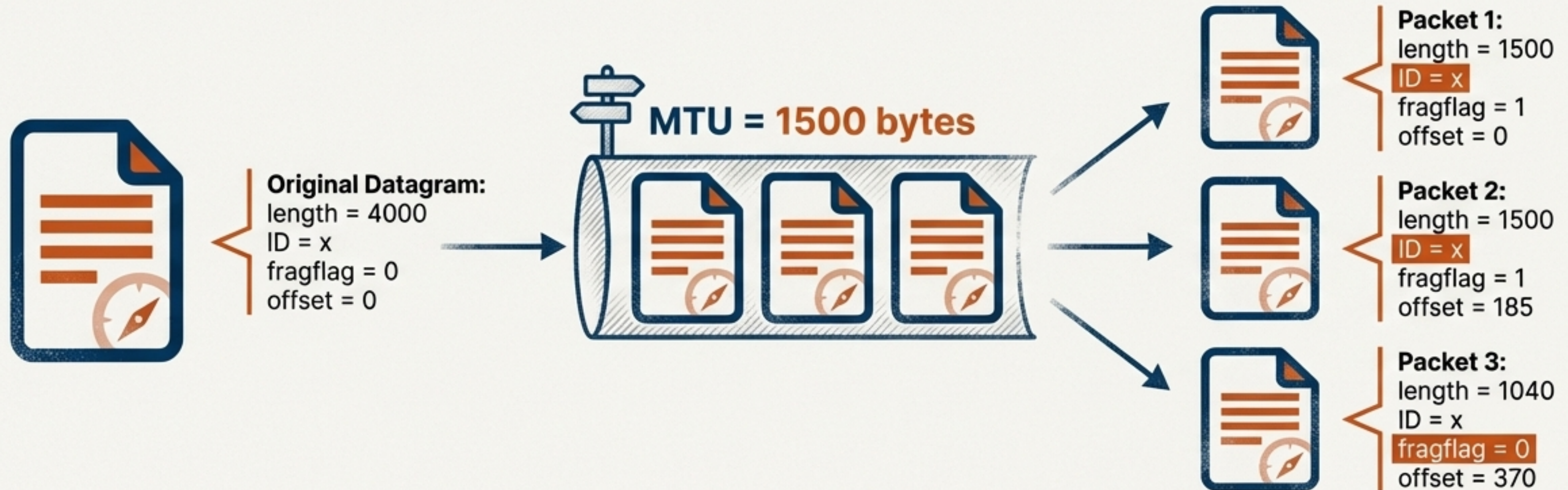
	Prefix Match	Output Link
21-bit match →	11001000 00010111 00010***	0
	11001000 00010111 00011***	2

Longest Match!
Forward to Link 0.

Roadblock Ahead: When the Path Narrows

Different network links have different size limits, called the Maximum Transfer Unit (MTU). If our datagram is larger than the next link's MTU, the router must **fragment** it into smaller pieces. Each piece becomes a new, smaller IP datagram. They are only reassembled at the final destination.

The original IP header's **identification**, **flags**, and **fragment offset** fields are used to track the pieces so they can be put back together correctly.



The Traveler's Guarantee: 'Best Effort' Service

What guarantees does our datagram have on its journey? Surprisingly few. The Internet's service model is called '**Best Effort**'. This simplicity is what allowed it to scale so successfully. Reliability is left as a job for higher-level protocols like TCP.



What We Promise

 **Simplicity & Scalability**
The core advantage that enables global connectivity and massive growth.

What We Don't

-  **Guaranteed Delivery**
Packets can be dropped due to router queue overflows (congestion).
-  **Guaranteed Order**
Packets can take different routes and arrive out of order.
-  **Guaranteed Timing**
Delays are variable and unpredictable.

A Shrinking World: The IPv4 Address Crisis

The **32-bit IPv4 address space** provides approximately **4.3 billion** unique addresses. In the 1970s and 80s, this seemed inexhaustible. Today, with billions of people and countless connected devices, this address space is effectively **exhausted**. While **NAT** provided a crucial short-term fix, a long-term solution was needed.

“Who the hell knew how much address space we needed?”
— Vint Cerf (reflecting on the decision to use 32-bit addresses)



The Next Generation: The IPv6 Superhighway

IPv6 is the long-term solution to address exhaustion and a chance to redesign the IP header for better performance.

Massive Address Space



IPv4: 200.23.16.1

IPv6: 2001:0db8:85a3:0000:0000:8a2e:0370:7334

128-bit addresses provide 3.4×10^{38} unique addresses. That's more than enough for every grain of sand on Earth to have its own.

Streamlined Header

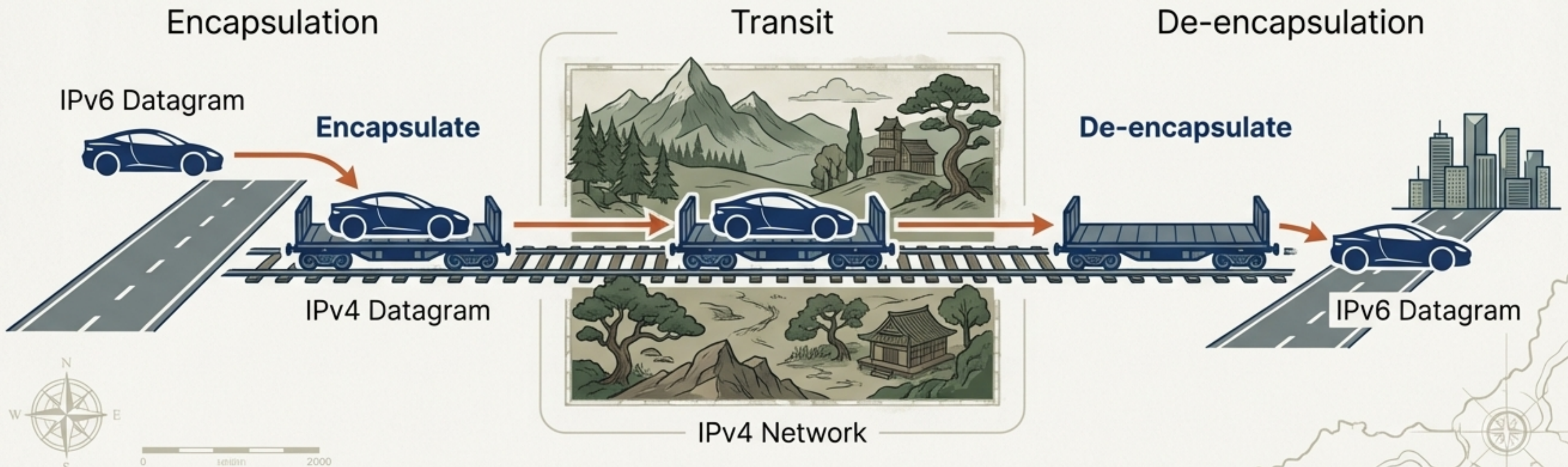
Version	Traffic Class	Flow Label	
Payload Length		Next Header	Hop Limit
Source Address			
Destination Address			

40 byte

- ✓ **No Checksum:** Higher layers are expected to handle error checking, removing a calculation from every router's workload.
- ✓ **No Fragmentation:** Routers no longer fragment packets. If a packet is too large, it's dropped, and the source is notified to send smaller packets.
- ✓ **No Options:** The complex 'options' field is removed, replaced by a more flexible 'next header' field.

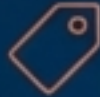
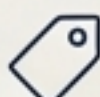
Crossing Borders: Tunneling Through the Old World

The world can't switch to IPv6 overnight. For now, the two protocols must coexist. When an IPv6 datagram needs to cross a part of the internet that only understands IPv4, it uses a technique called **tunneling**. The entire IPv6 datagram is encapsulated *inside* an IPv4 datagram. It becomes the payload of the IPv4 packet.



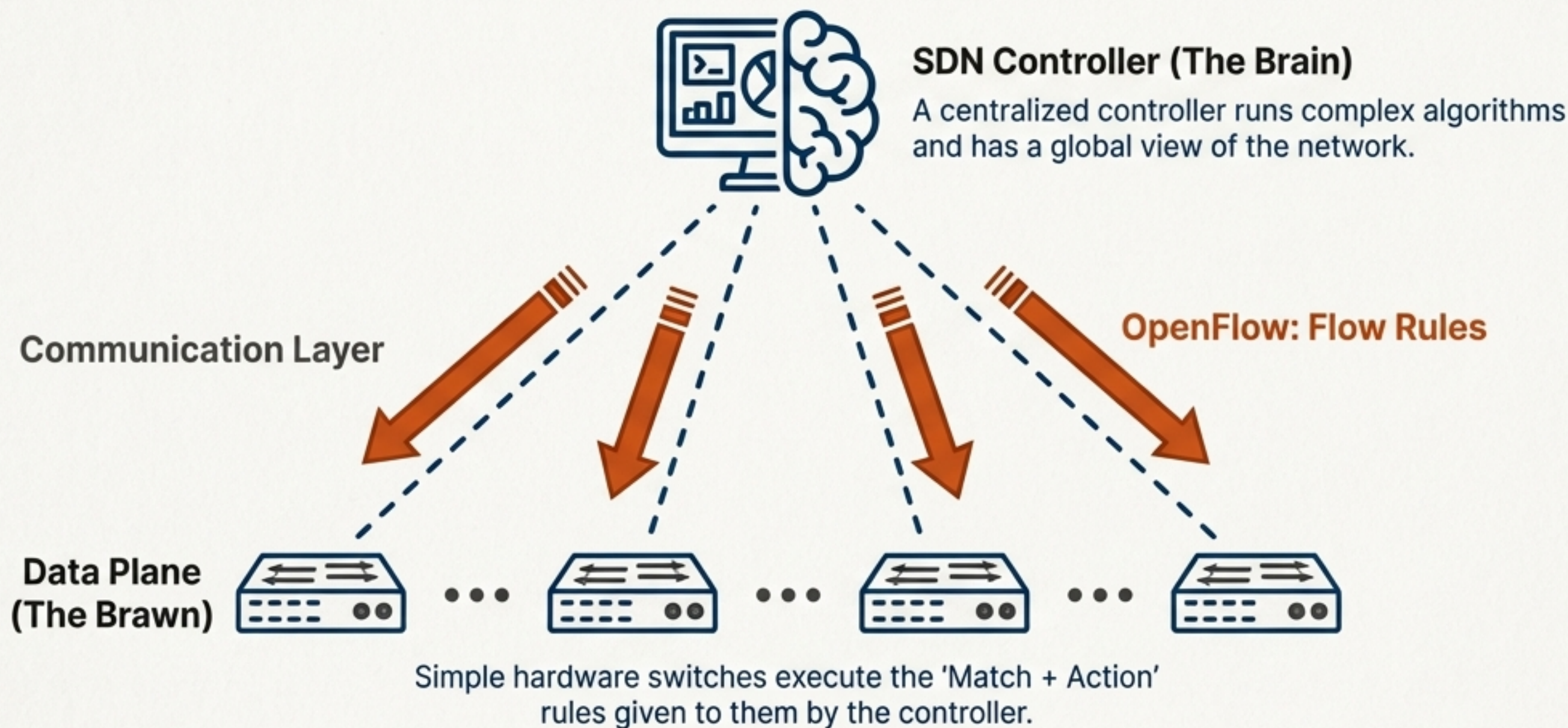
A Smarter Crossroads: The 'Match + Action' Revolution

Traditional forwarding is based on one field: the destination IP address. But what if a router could make decisions based on *any* header field, across multiple layers? This is the powerful abstraction of **Generalized Forwarding**.

 Match (Pattern)	 Action	 Use Case
 IP Dst = 51.6.0.8	 forward(6)	 Standard Destination-based Forwarding
 TCP Dst Port = 22	 drop	 Firewall: Block SSH Traffic
 IP Src = 10.1.2.3, TCP Src Port > 1024	 send_to_controller	 Monitoring: Analyze specific user traffic
 MAC Src = E4:54:E8:12:34:56	 modify(VLAN_ID=100), forward(3)	 QoS: Prioritize traffic from a VIP device

The Programmable Network: From Rules to Intelligence

The “**Match + Action**” concept is the foundation of **OpenFlow**, the protocol that enables Software-Defined Networking (SDN). SDN separates the brain from the brawn, making the network programmable from a central point.



The Journey's End: A Datagram's Tale

We followed a datagram across the internet, witnessing the key functions of the network layer's data plane. This journey, repeated trillions of times a day, is what makes our connected world possible.

